

Step-by-Step Hand Calculation of PC1 (Principal Component Analysis)

Dataset:

Study Hour: 8,7,6,5,9

Exam Score: 79,75,73,79,82

Step 1: Mean

Study Hour mean = $(8+7+6+5+9)/5 = 7$

Exam Score mean = $(79+75+73+79+82)/5 = 77.6$

Step 2: Center Data

Student	Study Hour (x)	Centered x = (x-7)	Calculation	Exam Score (y)	Centered y = (y-77.6)	Calculation
1	8	1	$8 - 7 = 1$	79	1.4	$79 - 77.6 = 1.4$
2	7	0	$7 - 7 = 0$	75	-2.6	$75 - 77.6 = -2.6$
3	6	-1	$6 - 7 = -1$	73	-4.6	$73 - 77.6 = -4.6$
4	5	-2	$5 - 7 = -2$	79	1.4	$79 - 77.6 = 1.4$
5	9	2	$9 - 7 = 2$	82	4.4	$82 - 77.6 = 4.4$

Step 3: Covariance Matrix

(a) Variance of Study Hour, Var(x)

Centered Study Hour values:

[1, 0, -1, -2, 2]

প্রতিটি মানের square করি।

Value	Square
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1	$1^2 = 1$
0	$0^2 = 0$
-1	$(-1)^2 = 1$
-2	$(-2)^2 = 4$
2	$2^2 = 4$

Sum of squares

$$1 + 0 + 1 + 4 + 4 = 10$$

Variance

$$\begin{aligned} Var(x) &= \frac{10}{5 - 1} \\ &= \frac{10}{4} \\ &= 2.5 \end{aligned}$$

(b) Variance of Exam Score, Var(y)

Centered Exam Score values:

$$[1.4, -2.6, -4.6, 1.4, 4.4]$$

Square of each value

Value	Square
1.4	$1.4^2 = 1.96$
-2.6	$(-2.6)^2 = 6.76$
-4.6	$(-4.6)^2 = 21.16$
1.4	$1.4^2 = 1.96$
4.4	$4.4^2 = 19.36$

Sum

$$\begin{aligned} & 1.96 + 6.76 + 21.16 + 1.96 + 19.36 \\ & = 51.20 \end{aligned}$$

Variance

$$\begin{aligned} Var(y) &= \frac{51.20}{5 - 1} \\ &= \frac{51.20}{4} \\ &= 12.8 \end{aligned}$$

(c) Covariance of x and y

প্রতিটি centered value pair গুণ করি।

x	y	Product
1	1.4	$1 \times 1.4 = 1.4$
0	-2.6	$0 \times (-2.6) = 0$
-1	-4.6	$(-1) \times (-4.6) = 4.6$
-2	1.4	$(-2) \times 1.4 = -2.8$
2	4.4	$2 \times 4.4 = 8.8$

Sum

$$\begin{aligned} & 1.4 + 0 + 4.6 - 2.8 + 8.8 \\ & = 12 \end{aligned}$$

Covariance

$$\begin{aligned} Cov(x, y) &= \frac{12}{5 - 1} \\ &= \frac{12}{4} \\ &= 3 \end{aligned}$$

Covariance Matrix

$$\Sigma = \begin{bmatrix} 2.5 & 3 \\ 3 & 12.8 \end{bmatrix}$$

Step 4: Calculate the Eigenvalues

Covariance Matrix

$$A = \begin{bmatrix} 2.5 & 3 \\ 3 & 12.8 \end{bmatrix}$$

Characteristic equation

$$\begin{aligned} |A - \lambda I| &= 0 \\ \begin{vmatrix} 2.5 - \lambda & 3 \\ 3 & 12.8 - \lambda \end{vmatrix} &= 0 \end{aligned}$$

Determinant expand করি

$$\begin{aligned} (2.5 - \lambda)(12.8 - \lambda) - (3 \times 3) &= 0 \\ (2.5 - \lambda)(12.8 - \lambda) - 9 &= 0 \end{aligned}$$

Multiply করি

$$\begin{aligned} 32 - 2.5\lambda - 12.8\lambda + \lambda^2 - 9 &= 0 \\ \lambda^2 - 15.3\lambda + 23 &= 0 \end{aligned}$$

Quadratic equation solve করলে

$$\begin{aligned} \lambda_1 &\approx 13.58 \\ \lambda_2 &\approx 1.72 \end{aligned}$$

Largest Eigenvalue (PC1)

$$\boxed{\lambda_1 = 13.58}$$

Step 5: Calculate the Eigenvector (PC1)

Largest eigenvalue

$$\lambda = 13.58$$

Equation

$$(A - \lambda I)v = 0$$

Substitute করি

$$\begin{bmatrix} 2.5 - 13.58 & 3 \\ 3 & 12.8 - 13.58 \end{bmatrix} \\ = \begin{bmatrix} -11.08 & 3 \\ 3 & -0.78 \end{bmatrix}$$

প্রথম row থেকে

$$\begin{aligned} -11.08x + 3y &= 0 \\ 3y &= 11.08x \\ y &= \frac{11.08}{3}x \\ y &= 3.693x \end{aligned}$$

ধরি,

$$x = 1$$

তাহলে

$$y = 3.693$$

Eigenvector

$$\begin{bmatrix} 1 \\ 3.693 \end{bmatrix}$$

Normalize the Eigenvector

Length

$$\begin{aligned} &= \sqrt{1^2 + 3.693^2} \\ &= \sqrt{1 + 13.638} \end{aligned}$$

$$= \sqrt{14.638}$$

$$= 3.826$$

Normalized Eigenvector

$$\left(\frac{1}{3.826}, \frac{3.693}{3.826} \right)$$

$$= (0.261, 0.965)$$

Final Principal Component (PC1)

$$\boxed{\text{PC1} = 0.261 \times \text{Study Hour} + 0.965 \times \text{Exam Score}}$$

এইটাই প্রথম Principal Component, যা **dataset-এর সর্বাধিক variance capture করে।**